

Acute effects of coffee on endothelial function in healthy subjects.

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BACKGROUND/OBJECTIVES: Coffee is the most widely consumed beverage in the world, but its effect on the cardiovascular system has not been fully understood. Coffee contains caffeine and antioxidants, which may influence endothelial function, both of which have not yet been investigated. The objective of this study was to investigate the acute effects of coffee on endothelial function measured by brachial artery flow-mediated dilation (FMD). SUBJECTS/METHODS: A total of 20 (10 males and 10 females) healthy non-obese subjects underwent a double-blind, crossover study. Subjects ingested one cup of caffeinated (CC) and one cup of decaffeinated (DC) Italian espresso coffee in random order at 5- to 7-day intervals. RESULTS: Following CC ingestion, FMD decreased progressively and significantly (mean $\pm$ s.e.m.: 0 min, 7.7 $\pm$ 0.6; 30 min, 6.3 $\pm$ 0.7; 60 min, 6.0 $\pm$ 0.8%; ANOVA (analysis of variance), $P<0.05$), but it did not significantly increase after DC ingestion (0 min, 6.9 $\pm$ 0.6; 30 min, 8.1 $\pm$ 0.9; 60 min, 8.5 $\pm$ 0.9%; $P=0.115$). Similarly, CC significantly increased both systolic and diastolic blood pressure; this effect was not observed after DC ingestion. Blood glucose concentrations remained unchanged after ingestion of both CC and DC, but insulin (0 min, 15.8 $\pm$ 0.9; 60 min, 15.0 $\pm$ 0.8 μ U/ml; $P<0.05$) and C-peptide (0 min, 1.25 $\pm$ 0.09; 60 min, 1.18 $\pm$ 0.09 ng/ml; $P<0.01$) blood concentrations decreased significantly only after CC ingestion. CONCLUSIONS: CC acutely induced unfavorable cardiovascular effects, especially on endothelial function. In the fasting state, insulin secretion is also likely reduced after CC ingestion. Future studies will determine whether CC has detrimental clinically relevant effects, especially in unhealthy subjects.